Bioinformatics: Hidden Markov Models

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Book Reference

Bioinformatics Algorithms, An Active Learning Approach , Vol 2, Chapter 10

HIV Virus

- In 1984, US Health and Human Services Secretary Margaret Heckler declared that an HIV vaccine would be available within two years, stating, "Yet another terrible disease is about to yield to patience, persistence and outright genius."
- In 1997, Bill Clinton established a new research center at the National Institutes of Health with the goal of developing an HIV vaccine. In his words, "It is no longer a question of whether we can develop an AIDS vaccine, it is simply a question of when."
- In 2005, Merck began clinical trials of an HIV vaccine but discontinued them two years later after learning that the vaccine actually increased the risk of HIV infection in some recipients.
- 40 million people suffering to this date - scientists have developed a successful **antiretroviral** therapy, a drug cocktail that stabilizes an infected patient's symptoms.

Traditional Vaccines

- Classical vaccines against viruses are often made from the surface proteins of a virus.
- These vaccines stimulate the human immune system to recognize viral envelope proteins as foreign, destroy them, and keep a record of it, so that the immune system can identify and eradicate the virus in a later encounter.
- Idea - create a single peptide that contains the least variable segments of the envelope proteins taken from all known HIV strains and use this peptide as the basis for a universal vaccine fighting all HIV strains.
- Bad news: HIV envelope proteins mutate fast, and **masked** by glycosylation, a post-translational modification that often makes these proteins invisible to the human immune system.

Mutating HIV

```
VKKLGEQFR-NKTIIFNQPSGGDLEIVMHSFNCGGEFFYCNTTQLFN----------NSTES------DTITL
VKKLGEQFR-NKTIIFNQPSGGDLEIVMHSFNCGGEFFYCNTTQLFN----------NSTDNG-----DTITL
VKKLGEQFR-NKTIIFNQPSGGDLEIVMHSFNCGGEFFYCNTTQLFD----------NSTESNN----DTITL
VDKLREQFGKNKTIIFNQPSGGDLEIVMHTFNCGGEFFYCNTTQLFNSTWNS---TGNGTESYNGQENGTITL
VDKLREQFGKNKTIIFNQPSGGDLEIVMHTFNCGGEFFYCNTTQLFNSTWNG---TNTT--GLDG--NDTITL
VDKLREQFGKNKTIIFNQSSGGDLEIVTHTFNCGGEFFYCNTTQLFNSNWTG---NSTE--GLHG--DDTITL
VKKLGEQFG-NKTIIFNQSSGGGLEIVMHSFNCGGEFFYCNTTQLFNN--TR-----NSTESNNGQGNDTTTL
VKKLREQFGKNKTIIFKQSSGGDLEIVTHTFNCAGEFFYCNTTQLFNSNWTE-----NSITGLDG--NDTITL
VGKLREQFGK-KTIIFNQPSGGDLEIVMHSFNCQGEFFYCNTTRLFNSTWDNSTWNSTGKDKENGN-NDTITL
```

FIGURE 10.1 A multiple alignment of a short region of gp120 proteins sampled from a single HIV-positive patient at nine different time points. Almost half of the columns (shown in darker text) are not conserved across all time points, illustrating how quickly HIV evolves, even within an individual host. Amino acids differing from the most common symbol in a column are shown in blue.

HIV Genes

- Rapidly mutating env gene, which has a mutation rate of 1 to 2% per nucleotide per year.
- env gene then gets cut into glycoprotein gp120 (approx. 480 amino acids) and glycoprotein gp41 (approx. 345 amino acids). Together, gp120 and gp41 form the envelope spike, which mediates entry of the HIV virus into human cells.
- During infection, viral proteins like gp120 that are used by HIV to enter the cell are transported to the cell surface, where they can cause the host cell membrane to fuse with neighboring cells. This causes dozens of human cells to fuse their cell membranes into a giant, nonfunctional **syncytium**, or abnormal multinucleae cell.
- This mechanism allows an SI virus to kill many human cells by infecting only one.

Mutating HIV

```
CMRPGNNTRKSIHMGPGKAFYATGDIIGDIRQAHC
CMRPGNNTRKSIHMGPGRAFYATGDIIGDTRQAHC
CMRPGNNTRKSIHIGPGRAFYATGDIIGDIRQAHC
CMRPGNNTRKSIHIGPGRAFYTTGDIIGDIRQAHC
CTRPNNNTRKGISIGPGRAFIAARKIIGDIRQAHC
CTRPNNYTRKGISIGPGRAFIAARKIIGDIRQAHC
CTRPNNNTRKGIRMGPGRAFIAARKIIGDIRQAHC
CVRPNNYTRKRIGIGPGRTVFATKQIIGNIRQAHC
CTRPSNNTRKSIPVGPGKALYATGAIIGNIRQAHC
CTRPNNHTRKSINIGPGRAFYATGEIIGDIRQAHC
CTRPNNNTRKSINIGPGRAFYATGEIIGDIRQAHC
CTRPNNNTRKSIHIGPGRAFYTTGEIIGDIRQAHC
CTRPNNNTRKSINIGPGRAFYTTGEIIGNIRQAHC
CIRPNNNTRGSIHIGPGRAFYATGDIIGEIRKAHC
CIRPNN-TRRSIHIGPGRAFYATGDIIGEIRKAHC
CTRPGSTTRRHIHIGPGRAFYATGNILGSIRKAHC
CTRPGSTTRRHIHIGPGRAFYATGNI-GSIRKAHC
CTGPGSTTRRHIHIGPGRAFYATGNIHG-IRKGHC
CMRPGNNTRRRIHIGPGRAFYATGNI-GNIRKAHC
CMRPGTTTRRRIHIGPGRAFYATGNI-GNIRKAHC
```

FIGURE 10.3 A multiple alignment of the V3 loop region taken from twenty HIV patients, and the motif logo of this alignment. The alignment's 11th and 25th columns are shown in darker text; occurrences of arginine (R) or lysine (K) in these columns are shown in red. The 11/25 rule will classify six of the patients as infected with an SI isolate. Although the V3 loop is an important and rather conserved segment of gp120, the level of conservation varies across different positions. For example, whereas the first and last positions are extremely conserved, positions 11 and 25 exhibit high variation.

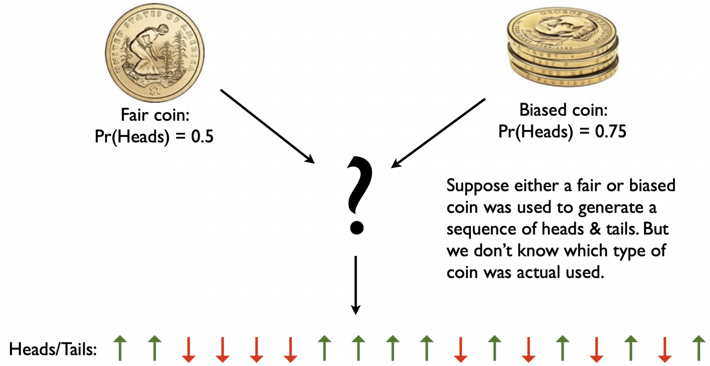
Multiple Sequence Alignment

- Why using the same amino acid scoring matrix (as well as indel penalties) across different columns of an alignment?
- Solution? Profile HMM

Finding CG Islands

- Different species have widely varying GC-content, or the percentage of cytosine and guanine nucleotides in a genome. For example, the human genome's GC-content is approximately 42%.
- After accounting for the human genome's skewed GC-content, we might expect that each of the dinucleotides CC, CG, GC, and GG would occur in the human genome with frequency $0.21 \cdot 0.21 = 4.41\%$.
- However, the frequency of CG in the human genome is only about 1%! This dinucleotide is so rare because of methylation, the most common DNA modification, which typically adds a methyl group (CH3) to the cytosine nucleotide within a CG dinucleotide. The resulting methylated cytosine has the tendency to further deaminate into thymine
- Methylation is often suppressed around genes in areas called CG-islands, where CG appears relatively frequently

- How could we guess which coin was more likely?



- Compute the Probability of the Observed Sequence

Fair coin: Pr(Heads) = 0.5
Biased coin: Pr(Heads) = 0.75

x = ↑ ↑ ↓ ↓ ↓ ↓ ↑

$Pr(x \mid Fair) = 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 = 0.5^7 = 0.0078125$

$Pr(x \mid Biased) = 0.75 \times 0.75 \times 0.25 \times 0.25 \times 0.25 \times 0.25 \times 0.75 = 0.001647949$

The *log-odds score*:

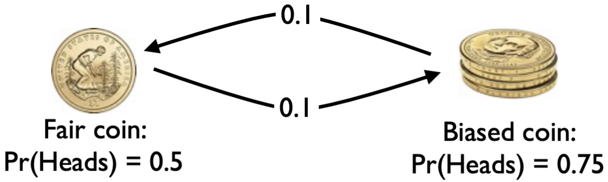
$$\log_2 \frac{Pr(x \mid Fair)}{Pr(x \mid Biased)} = \log_2 \frac{0.0078}{0.0016} = 2.245$$

> 0. Hence "Fair" is a better guess.

Casino

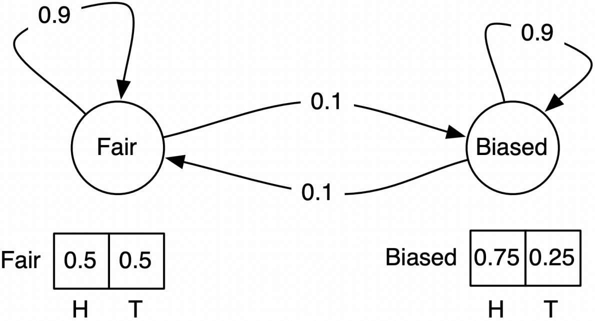
- What if the casino switches coins?
- How can we compute the probability of the entire sequence?
- How could we guess which coin was more likely at each position?

Fair coin: Pr(Heads) = 0.5
Biased coin: Pr(Heads) = 0.75
Probability of switching coins = 0.1



Hidden Markov Model

Fair coin: Pr(Heads) = 0.5
Biased coin: Pr(Heads) = 0.75
Probability of switching coins = 0.1



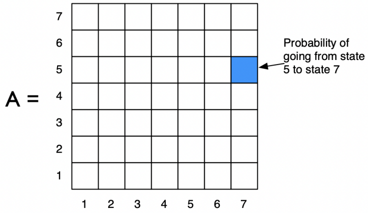
Hidden Markov Model

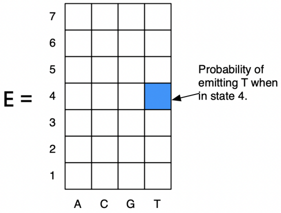
Σ = alphabet of symbols.

Q = set of states.

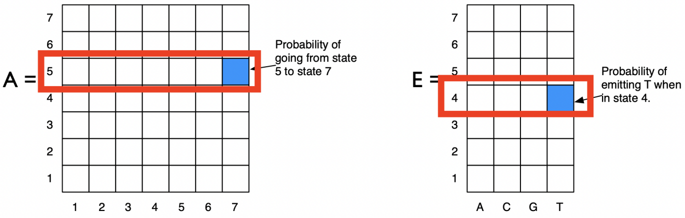
A = an |Q| x |Q| matrix where entry (k,l) is the probability of moving from state k to state l.

E = a |Q| x |Σ| matrix, where entry (k,b) is the probability of emitting b when in state k.



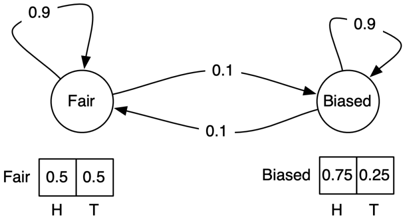


Hidden Markov Model



Sum of the # in each row must be 1.

Hidden Markov Model



x = ↓ ↑ ↓ ↑ ↑ ↑ ↓ ↑ ↑ ↓

π = F F F B B B B F F F

The decoding problem

Decoding Problem:
Find an optimal hidden path in an HMM given a string of its emitted symbols.

Input: A string $x = x_1 \ldots x_n$ emitted by an HMM (Σ, *States*, *Transition*, *Emission*).
Output: A path π that maximizes the probability $\Pr(x, \pi)$ over all possible paths through this HMM.

$$
\begin{aligned}
\Pr(x,\pi) &= \Pr(x|\pi) \cdot \Pr(\pi) \\
&= \prod_{i=1}^{n} \Pr(x_i|\pi_i) \cdot \Pr(\pi_{i-1} \rightarrow \pi_i) \\
&= \prod_{i=1}^{n} \mathit{emission}_{\pi_i}(x_i) \cdot \mathit{transition}_{\pi_{i-1},\pi_i} \,.
\end{aligned}
$$

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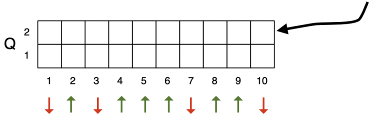
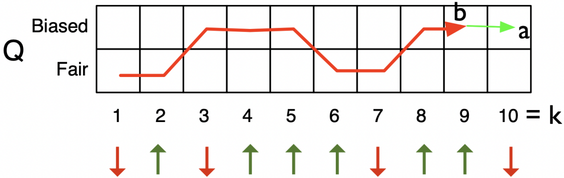
A[a, k] := the probability of the **best** path for $x_1 \ldots x_k$ that ends at state a.



A[a, k] = the path for $x_1 \ldots x_{k-1}$ that goes to some state *b* times cost of a transition from b to i, and then to output x_k from state a.



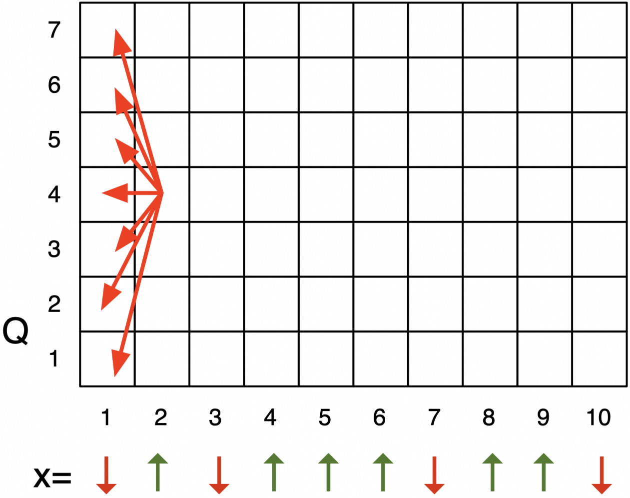
Viterbi DP Recurrence

$$A[a,k] = \underbrace{\max_{b \in Q}}_{\substack{\text{Over all possible} \\ \text{previous states.}}} \{\underbrace{A[b,k-1]}_{\substack{\text{Best path for} \\ x_1..x_k \text{ ending} \\ \text{in state } b}} \times \underbrace{\Pr(b \to a)}_{\substack{\text{Probability of} \\ \text{transitioning} \\ \text{from state } b \text{ to} \\ \text{state } a}} \times \underbrace{\Pr(x_k \mid \pi_k = a)}_{\substack{\text{Probability of} \\ \text{outputting } x_k \\ \text{given that the} \\ k\text{th state is } a.}}\}$$

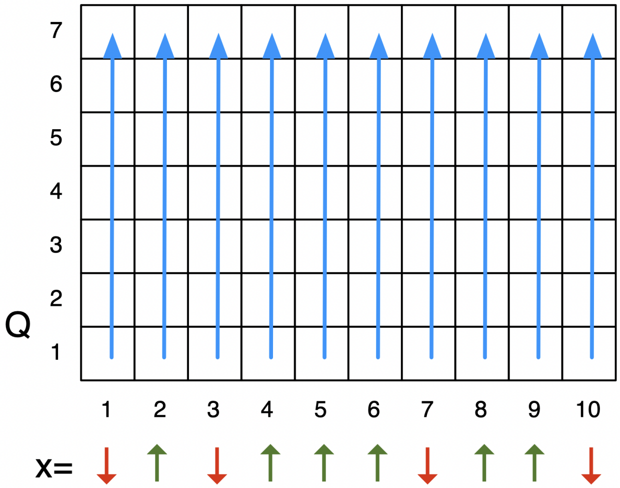
Base case:

$$A[a,1] = \underbrace{\Pr(\pi_1 = a)}_{\substack{\text{Probability} \\ \text{that the first} \\ \text{state is } a}} \times \underbrace{\Pr(x_1 \mid \pi_1 = a)}_{\substack{\text{Probability of} \\ \text{emitting } x_1 \\ \text{given the first} \\ \text{state is } a.}}$$

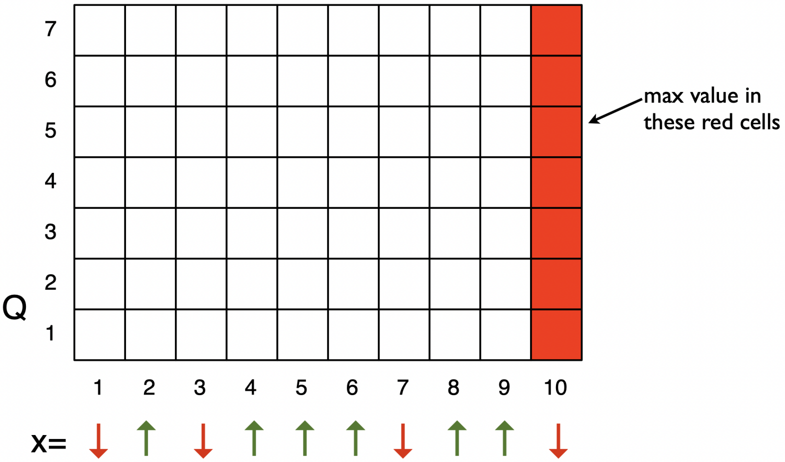
Which Cells Do We Depend On?



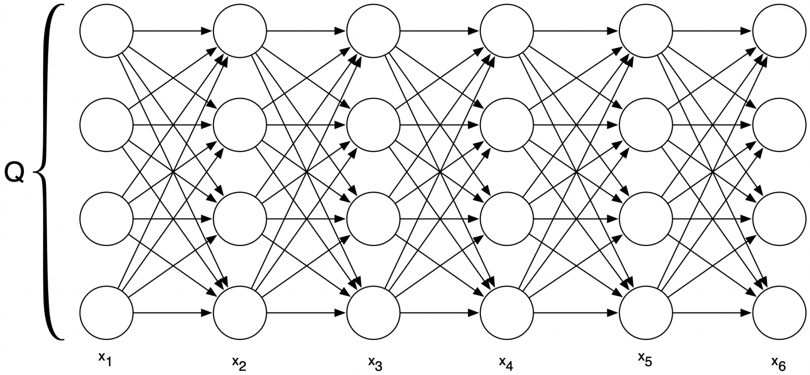
Order to Fill in the Matrix:



Where's the answer?



Graph view of viterbi



- Runtime? $O(n|Q|^2)$

Using Logs

Typically, we take the log of the probabilities to avoid multiplying a lot of terms:

$$\begin{aligned}\log(A[a,k]) &= \max_{b\in Q}\{\log(A[b,k-1]\times \Pr(b\to a)\times \Pr(x_k \mid \pi_k = a))\}\\ &= \max_{b\in Q}\{\log(A[b,k-1]) + \log(\Pr(b\to a)) + \log(\Pr(x_k \mid \pi_k = a))\}\end{aligned}$$

$$\boxed{\textbf{Remember: } \log(ab) = \log(a) + \log(b)}$$

Why do we want to avoid multiplying lots of terms?

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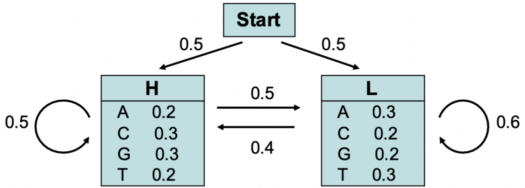
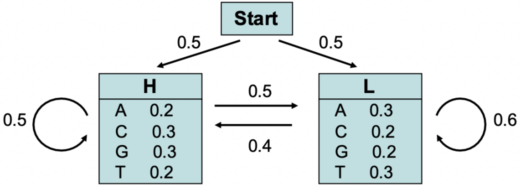
Viterbi - An example of decoding



Let's consider the following simple HMM. This model is composed of 2 states, **H** (high GC content) and **L** (low GC content). We can for example consider that state H characterizes coding DNA while L characterizes non-coding DNA.

The model can then be used to predict the region of coding DNA from a given sequence.

Viterbi - An example of decoding



Consider the sequence S= **GGCACTGAA**

There are several paths through the hidden states (H and L) that lead to the given sequence S.

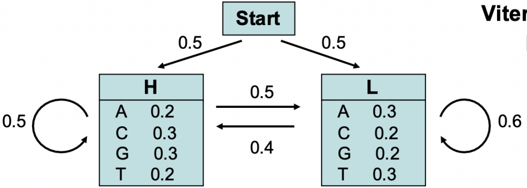
Example: P = **LLHHHHLLL**

The probability of the HMM to produce sequence S through the path P is:

$$p = p_L(0) * p_L(G) * p_{LL} * p_L(G) * p_{LH} * p_H(C) * \ldots$$
$$= 0.5 * 0.2 * 0.6 * 0.2 * 0.4 * 0.3 * \ldots$$
$$= \ldots$$

Viterbi - An example of decoding

Viterbi algorithm: principle



G G C $\boxed{\text{A}}$ C T G A A

The probability of the most probable path ending in state **k** with observation "i" is

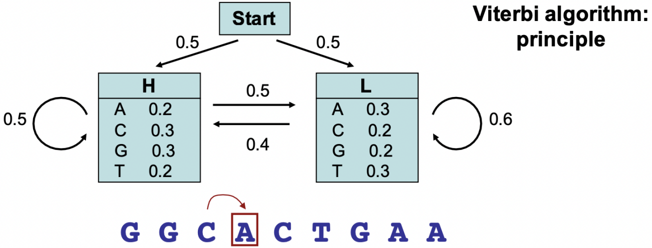
$$p_l(i,x) = e_l(i)\max_k\left(p_k(j,x-1).p_{kl}\right)$$

probability to observe element *i* in state *l*

probability of the most probable path ending at position x-1 in state *k* with element *j*

probability of the transition from state *l* to state *k*

Viterbi - An example of decoding



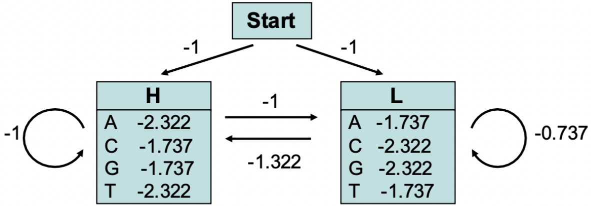
The probability of the most probable path ending in state **k** with observation "i" is

$$p_l(i,x) = e_l(i)\max_k\left(p_k(j,x-1).p_{kl}\right)$$

In our example, the probability of the most probable path ending in state **H** with observation "A" at the 4th position is:

$$p_H(A,4) = e_H(A)\max\left(p_L(C,3)p_{LH}, p_H(C,3)p_{HH}\right)$$

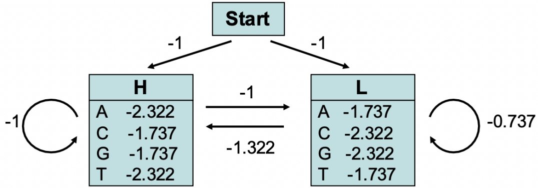
We can thus compute recursively (from the first to the last element of our sequence) the probability of the most probable path.



Remark: for the calculations, it is convenient to use the log of the probabilities (rather than the probabilities themselves). Indeed, this allows us to compute *sums* instead of *products*, which is more efficient and accurate.

We used here $\log_2(p)$.

Viterbi - An example of decoding



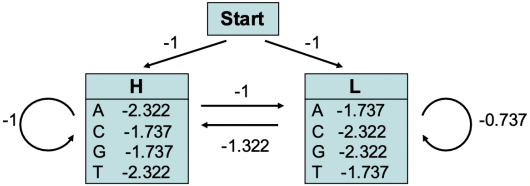
GGCACTGAA

Probability (in $\log_2$) that G at the first position was emitted by state **H**

$p_H(G,1) = -1 -1.737 = -2.737$

Probability (in $\log_2$) that G at the first position was emitted by state **L**

$p_L(G,1) = -1 -2.322 = -3.322$



GGCACTGAA

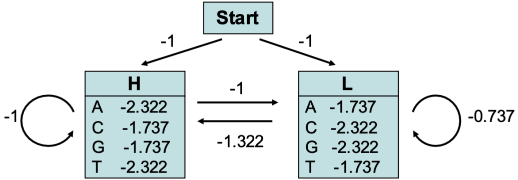
Probability (in $\log_2$) that G at the 2nd position was emitted by state **H**

$p_H(G,2) = -1.737 + \max(p_H(G,1)+p_{HH},\ p_L(G,1)+p_{LH})$
$= -1.737 + \max(-2.737 - 1\ ,\ -3.322 - 1.322)$
$= -5.474$ (obtained from $p_H(G,1)$)

Probability (in $\log_2$) that G at the 2nd position was emitted by state **L**

$p_L(G,2) = -2.322 + \max(p_H(G,1)+p_{HL},\ p_L(G,1)+p_{LL})$
$= -2.322 + \max(-2.737 - 1.322\ ,\ -3.322 - 0.737)$
$= -6.059$ (obtained from $p_H(G,1)$)

Viterbi - An example of decoding



GGCACTGAA

	G	G	C	A	C	T	G	A	A
H	-2.73	-5.47	-8.21	-11.53	-14.01	...			-25.65
L	-3.32	-6.06	-8.79	-10.94	-14.01	...			-24.49

We then compute iteratively the probabilities $p_H(i,x)$ and $p_L(i,x)$ that nucleotide *i* at position *x* was emitted by state **H** or **L**, respectively. The highest probability obtained for the nucleotide at the last position is the probability of the most probable path. This path can be retrieved by back-tracking.

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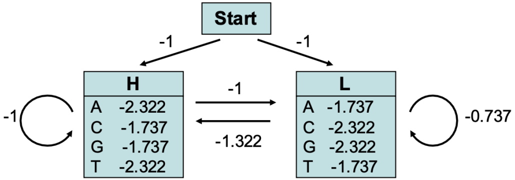

GGCACTGAA

back-tracking
(= finding the path which corresponds to the highest probability, -24.49)

	G	G	C	A	C	T	G	A	A
H	**-2.73** →	**-5.47** →	**-8.21** →	-11.53	-14.01	...			-25.65
L	-3.32	-6.06	-8.79	**-10.94** →	**-14.01** →	... →	→	→	→ **-24.49**

The most probable path is: **HHHLLLLLL**

Its probability is $2^{-24.49} = 4.25E-8$
(remember that we used $\log_2(p)$)

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Outcome Likelihood

- What is Pr(x), the probability that the HMM will emit a string x?
- Pr(x) is equal to the sum of $Pr(x, \pi)$ over all hidden paths π.
- We denote the total product weight of all paths from source to node (k, i) in the Viterbi graph as $forward_{k,i}$

$$Pr(x) = \sum_{\text{all states } k} forward_{k,n}$$

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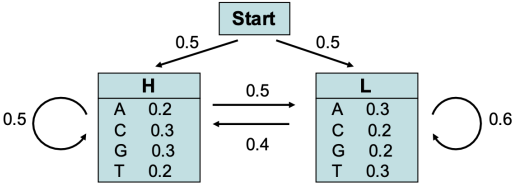
Forward Algorithm - An example



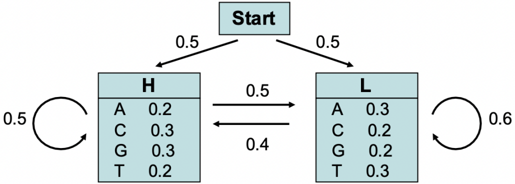
Consider now the sequence S= **GGCA**

What is the probability P(S) that this sequence S was generated by the HMM model?

This probability P(S) is given by the sum of the probabilities $p_i(S)$ of each possible path that produces this sequence.

The probability P(S) can be computed by dynamical programming using either the so-called **Forward** or the **Backward** algorithm.

Forward Algorithm - An example

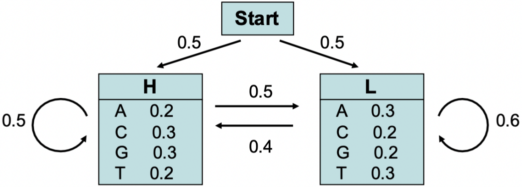


Forward algorithm

Consider now the sequence S= **GGCA**

	Start	G	G	C	A
H	0	0.5*0.3=0.15			
L	0	0.5*0.2=0.1			

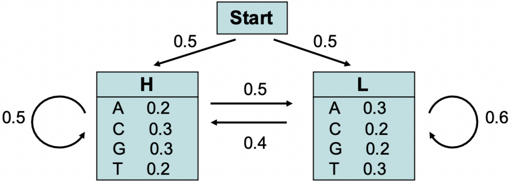
Forward Algorithm - An example



Consider now the sequence S= **GGCA**

	Start	G	G	C	A
H	0	0.5*0.3=0.15	0.15*0.5*0.3 + 0.1*0.4*0.3=0.0345		
L	0	0.5*0.2=0.1			

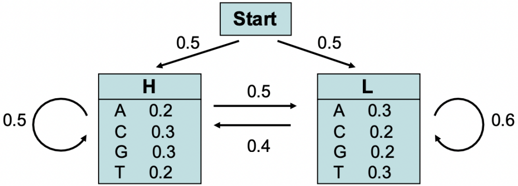
Forward Algorithm - An example



Consider now the sequence S= **GGCA**

	Start	G	G	C	A
H	0	0.5*0.3=0.15	0.15*0.5*0.3 + 0.1*0.4*0.3=0.0345		
L	0	0.5*0.2=0.1	0.1*0.6*0.2 + 0.15*0.5*0.2=0.027		

Forward Algorithm - An example



Consider now the sequence S= **GGCA**

	Start	G	G	C	A
H	0	0.5*0.3=0.15	0.15*0.5*0.3 + 0.1*0.4*0.3=0.0345	... + ...	
L	0	0.5*0.2=0.1	0.1*0.6*0.2 + 0.15*0.5*0.2=0.027	... + ...	

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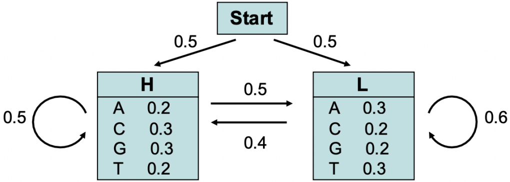
Forward Algorithm - An example



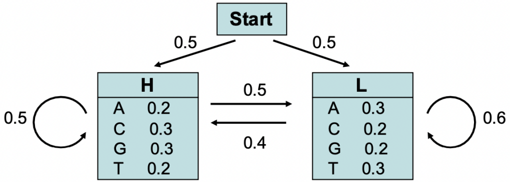
Consider now the sequence S= **GGCA**

	Start	G	G	C	A
H	0	0.5*0.3=0.15	0.15*0.5*0.3 + 0.1*0.4*0.3=0.0345	... + ...	0.0013767
L	0	0.5*0.2=0.1	0.1*0.6*0.2 + 0.15*0.5*0.2=0.027	... + ...	0.0024665

Σ = 0.0038432

=> The probability that the sequence S was generated by the HMM model is thus P(S)=0.0038432.

Forward Algorithm - An example



The probability that sequence S="GGCA" was generated by the HMM model is $P_{HMM}(S)$ = 0.0038432.

To assess the significance of this value, we have to compare it to the probability that sequence S was generated by the background model (i.e. by chance).

Ex: If all nucleotides have the same probability, $p_{bg}=0.25$; the probability to observe S by chance is: $P_{bg}(S) = p_{bg}^4 = 0.25^4 = 0.00396$.

Thus, for this particular example, it is likely that the sequence S does not match the HMM model ($P_{bg} > P_{HMM}$).

NB: Note that this toy model is very simple and does not reflect any biological motif. If fact both states H and L are characterized by probabilities close to the background probabilities, which makes the model not realistic and not suitable to detect specific motifs.

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Estimating HMM Parameters

$$(x^{(1)},\pi^{(1)}) = \begin{matrix} x_1^{(1)} x_2^{(1)} x_3^{(1)} x_4^{(1)} x_5^{(1)} \dots x_n^{(1)} \\ \pi_1^{(1)} \pi_2^{(1)} \pi_3^{(1)} \pi_4^{(1)} \pi_5^{(1)} \dots \pi_n^{(1)} \end{matrix}$$

$$(x^{(2)},\pi^{(2)}) = \begin{matrix} x_1^{(2)} x_2^{(2)} x_3^{(2)} x_4^{(2)} x_5^{(2)} \dots x_n^{(2)} \\ \pi_1^{(2)} \pi_2^{(2)} \pi_3^{(2)} \pi_4^{(2)} \pi_5^{(2)} \dots \pi_n^{(2)} \end{matrix}$$

Training examples where outputs and paths are known.

# of times transition $a \rightarrow b$ is observed.

# of times x was observed to be output from state a.

$$\Pr(a \rightarrow b) = \frac{A_{ab}}{\sum_{q \in Q} A_{aq}} \qquad \Pr(x \mid a) = \frac{E_{xa}}{\sum_{x \in \Sigma} E_{xq}}$$

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Pseudocounts

of times transition
$a \rightarrow b$ is observed.

# of times x was
observed to be
output from state a.

$$\Pr(a \to b) = \frac{A_{ab}}{\sum_{q \in Q} A_{aq}} \qquad \Pr(x \mid a) = \frac{E_{xa}}{\sum_{x \in \Sigma} E_{xq}}$$

What if a transition or emission is never observed in the training data?
$\Rightarrow$ 0 probability

Meaning that if we observe an example with that transition or emission in the real world, we will give it 0 probability.

But it's unlikely that our training set will be large enough to observe *every* possible transition.

Hence: we take A_{ab} = (#times a $\rightarrow$ b was observed) + 1 ← "pseudocount"
Similarly for E_{xa}.

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Viterbi Training - Baum Welch Algorithm

- **Problem**: typically, in the real would we only have examples of the output x, and we don't know the paths π.

Viterbi Training Algorithm:

1. Choose a random set of parameters.
2. Repeat:
   1. Find the best paths.
   2. Use those paths to estimate new parameters.

This is an local search algorithm.

It's also an example of a "Gibbs sampling" style algorithm.

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A first HMM for Multiple Sequence Alignment, θ

Alignment

1	2	3	4	5			6	7	8
A	C	D	E	F	A	C	A	D	F
A	F	D	A	–	–	–	C	C	F
A	–	–	E	F	D	–	F	D	C
A	C	A	E	F	–	–	A	–	C
A	D	D	E	F	A	A	A	D	F

*Alignment**

1	2	3	4	5	6	7	8
A	C	D	E	F	A	D	F
A	F	D	A	–	C	C	F
A	–	–	E	F	F	D	C
A	C	A	E	F	A	–	C
A	D	D	E	F	A	D	F

PROFILE(*Alignment**)

	1	2	3	4	5	6	7	8
A	1	0	0	1/5	0	3/5	0	0
C	0	2/4	0	0	0	1/5	1/4	2/5
D	0	1/4	3/4	0	0	0	3/4	0
E	0	0	0	4/5	0	0	0	0
F	0	1/4	0	0	1	1/5	0	3/5

$$M_1 \to M_2 \to M_3 \to M_4 \to M_5 \to M_6 \to M_7 \to M_8$$

Profile Probability

- Q1: What are the transition probabilities?
- Q2: What are the emission probabilities?
- The similarity score between Alignment* and Text is the probability Pr(Text) that the HMM for Alignment* emits Text.

	A	C	D	E	F	A	D	F
	A	F	D	A	-	C	C	F
*Alignment**	A	-	-	E	F	F	D	C
	A	C	A	E	F	A	-	C
	A	D	D	E	F	A	D	F
Text	A	D	D	A	F	F	D	F
emission probability	1	1/4	3/4	1/5	1	1/5	3/4	3/5

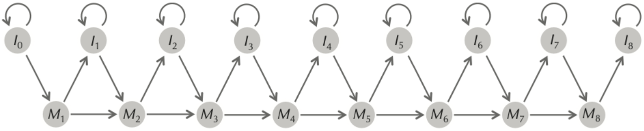


FIGURE 10.11 An HMM diagram for the seed alignment in Figure 10.9 with match and insertion states, abbreviated as M and I, respectively. The states I_0 and I_8 model insertions of symbols occurring before the beginning and end of *Alignment**, respectively.

Modelling Deletion

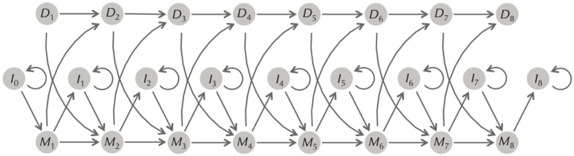
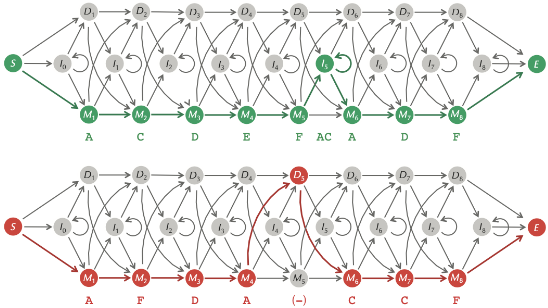


FIGURE 10.13 Adding silent deletion states (abbreviated as D_i) to the profile HMM diagram.

Profile HMM - Example

Alignment

A	C	D	E	F	AC	A	D	F
A	F	D	A	–	––	C	C	F
A	–	–	E	F	D–	F	D	C
A	C	A	E	F	––	A	–	C
A	D	D	E	F	AA	A	D	F



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Profile HMM

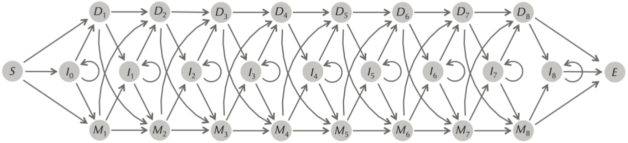
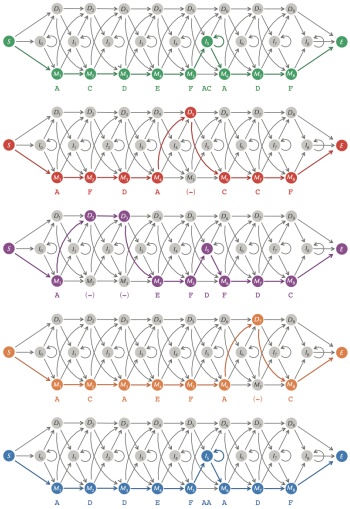


FIGURE 10.14 Adding transitions from insertion states to deletion states and vice-versa completes the profile HMM diagram for the profile matrix from Figure 10.9. Silent initial and terminal states are shown by S and E, respectively.

Profile HMM



Profile HMM

	S	I_0	M_1	D_1	I_1	M_2	D_2	I_2	M_3	D_3	I_3	M_4	D_4	I_4	M_5	D_5	I_5	M_6	D_6	I_6	M_7	D_7	I_7	M_8	D_8	I_8	E
S			1																								
I_0																											
M_1						.8	.2																				
D_1																											
I_1																											
M_2									1																		
D_2										1																	
I_2																											
M_3												1															
D_3												1															
I_3																											
M_4															.8	.2											
D_4																											
I_4																											
M_5																	.75	.25									
D_5																		1									
I_5																	.4	.6									
M_6																					.8	.2					
D_6																											
I_6																											
M_7																								1			
D_7																											
I_7																								1			
M_8																											1
D_8																											
I_8																											
E																											

FIGURE 10.16 The 27×27 matrix of transition probabilities for HMM(*Alignment*, 0.35), where *Alignment* is the multiple alignment from Figure 10.9. All values in empty cells are equal to zero. Cells shaded gray correspond to edges in the HMM diagram from Figure 10.14; cells shaded white correspond to forbidden transitions.

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Application

- Given a protein family, represented by Alignment, we can now return to the problem of deciding whether a newly sequenced protein, represented by Text, belongs to the family.
- To find the "best" alignment of Text against Alignment, we simply need to apply the Viterbi algorithm to find an optimal hidden path in HMM(Alignment, θ).
- If the product weight of this optimal hidden path exceeds a predetermined threshold, then we may conclude that Text belongs to the protein family, in which case we augment the existing seed alignment with an additional row corresponding to Text.
- How many rows and columns does the viterbi have?

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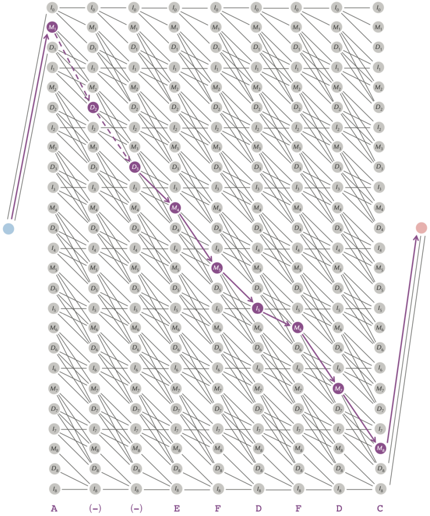
Wrong Viterbi



Right Viterbi

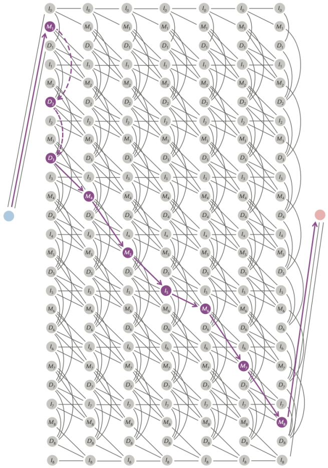


FIGURE 10.20 The Viterbi graph with |*States*| rows and |*Text*| columns for the profile HMM from Figure 10.14 emitting a string *Text* of length 7 so that edges entering deletion states are drawn downward within the same column instead of between columns as in Figure 10.18 and Figure 10.19. The purple path corresponds to the path through the HMM in Figure 10.15 emitting **AEFDFDC**.

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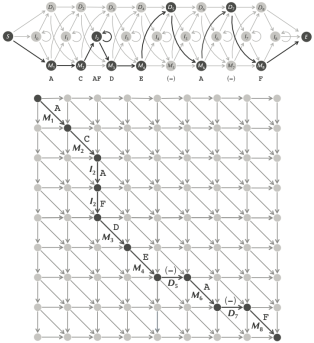

FIGURE 10.23 (Top) The hidden path through the profile HMM in Figure 10.17 (top) emitting ACAFDEAF. (Bottom) The path through a Manhattan-like graph corresponding to this hidden path.

$$s_{\text{MATCH}(j),i} = \max \begin{cases} s_{\text{MATCH}(j-1),i-1} \cdot \text{WEIGHT}_i(\text{MATCH}(j-1), \text{MATCH}(j)) \\ s_{\text{INSERTION}(j-1),i-1} \cdot \text{WEIGHT}_i(\text{INSERTION}(j-1), \text{INSERTION}(j)) \\ s_{\text{DELETION}(j-1),i-1} \cdot \text{WEIGHT}_i(\text{DELETION}(j-1), \text{DELETION}(j)) \end{cases}$$

A Question

STOP and Think: If the Viterbi algorithm for the crooked casino emits a path $\pi = \pi_1 \pi_2 \ldots \pi_n$ with $\pi_i = B$, is the dealer more likely to have used a biased coin at step i? Is it possible that $\pi_i = B$ but that $\Pr(\pi_i = B|x)$ is smaller than $\Pr(\pi_i = F|x)$?

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Soft Decoding

Soft Decoding Problem:
Find the probability that an HMM was in a particular state at a particular moment given its emitted string.

Input: A string $x = x_1 \dots x_n$ emitted by an HMM.
Output: The conditional probability $\Pr(\pi_i = k|x)$ that the HMM was in state k at step i given that it emitted x.

- The unconditional probability that a hidden path will pass through state k at time i and emit x can be written as the sum

$$Pr(\pi = k, x) = \sum_{\text{all paths } \pi \text{ with } \pi_i = k} Pr(x, \pi)$$

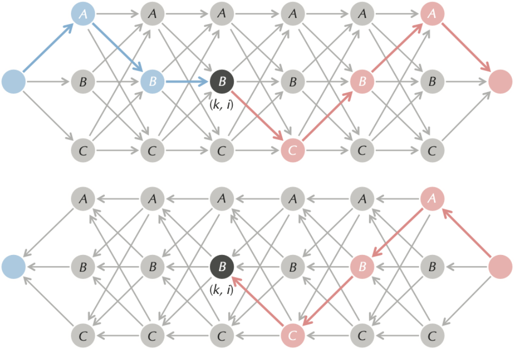

FIGURE 10.24 (Top) Each path from *source* to *sink* passing through the (black) node (k, i) in the Viterbi graph from Figure 10.7 (bottom) can be partitioned into two subpaths, one from *source* to (k, i) (shown in blue) and another from (k, i) to *sink* (shown in red). (Bottom) A "reversed Viterbi graph" in which all edges have been reversed, with a path from sink to (k, i) highlighted in red. The recurrence for $backward_{k,i}$ is based on computing $backward_{l,i+1}$ for every state l.

Problem Formulation

$$
\begin{aligned}
\Pr(\pi_i = k, x) &= \sum_{\text{all paths } \pi \text{ with } \pi_i = k} \Pr(x, \pi) \\
&= \sum_{\text{all paths } \pi_{\text{blue}}} \sum_{\text{all paths } \pi_{\text{red}}} \textsc{Weight}(\pi_{\text{blue}}) \cdot \textsc{Weight}(\pi_{\text{red}}) \\
&= \sum_{\text{all paths } \pi_{\text{blue}}} \textsc{Weight}(\pi_{\text{blue}}) \cdot \sum_{\text{all paths } \pi_{\text{red}}} \textsc{Weight}(\pi_{\text{red}}) .
\end{aligned}
$$

$$
\Pr(\pi_i = k, x) = \mathit{forward}_{k,i} \cdot \mathit{backward}_{k,i} .
$$

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